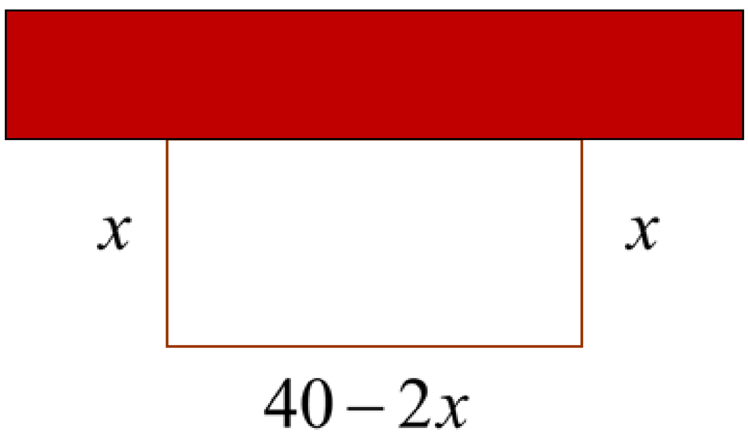
# Modeling and Optimization

## A Classic Problem

You have 40 feet of fence to enclose a rectangular garden along the side of a barn. What is the maximum area that you can enclose?




$$A = x(40-2x)$$

$$A = 40x - 2x^2$$

$$A' = 40 - 4x$$

$$0 = 40 - 4x$$

$$4x = 40$$

$$x = 10$$

There must be a local maximum here, since the endpoints are minimums.

$w = x$

$l = 40-2x$

$w = 10\text{ ft}$

$l = 20\text{ ft}$

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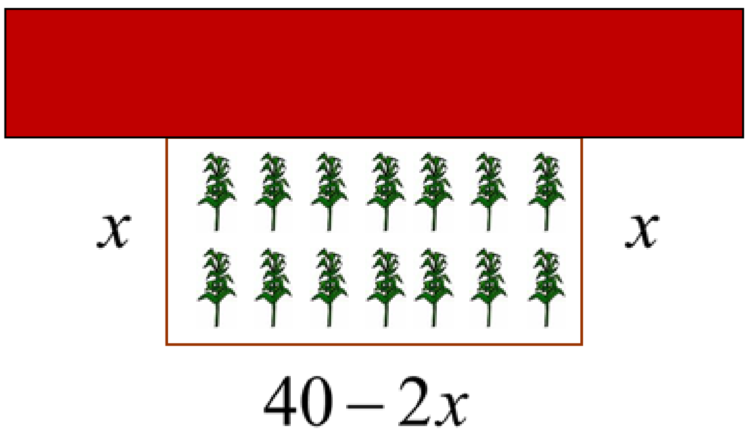



$$A = x(40-2x)$$

$$A = 40x - 2x^2$$

$$A' = 40 - 4x$$

$$0 = 40 - 4x$$

$$4x = 40$$

$$x = 10$$

$$A = 10(40 - 2\cdot 10)$$

$$A = 10(20)$$

$$\boxed{A = 200 \text{ ft}^2}$$

$$w = x$$

$$l = 40 - 2x$$

$$\boxed{\begin{array}{l} w = 10 \text{ ft} \\ l = 20 \text{ ft} \end{array}}$$

To find the maximum (or minimum) value of a function:

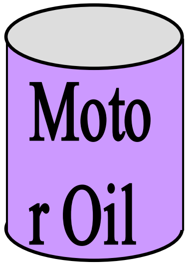
1. Write it in terms of <u>one</u> variable.

2. Find the first derivative and set it equal to zero.

3. Check the end points if necessary.

Example 5:

What dimensions for a one liter cylindrical can will use the least amount of material?




We can minimize the material by minimizing the area.

We need another equation that relates $r$ and $h$:

$$V = \pi r^2 h$$

$$\left(1\ \mathrm{L} = 1000\ \mathrm{cm}^3\right)$$

$$1000 = \pi r^2 h$$

$$\frac{1000}{\pi r^2} = h$$

$$A = \underbrace{2\pi r^2}_{\text{area of ends}} + \underbrace{2\pi r h}_{\text{lateral area}}$$

$$A = 2\pi r^2 + 2\cancel{\pi}\cancel{r} \cdot \frac{1000}{\cancel{\pi} \cancel{r^2}}$$

$$A = 2\pi r^2 + \frac{2000}{r}$$

$$A' = 4\pi r - \frac{2000}{r^2}$$

Example 5:

What dimensions for a one liter cylindrical can will use the least amount of material?

$$V = \pi r^2 h$$

$$\left(1\text{ L} = 1000\text{ cm}^3\right)$$

$$1000 = \pi r^2 h$$

$$\frac{1000}{\pi r^2} = h$$

$$\frac{1000}{\pi(5.42)^2} \approx h$$

$$\boxed{h \approx 10.83\text{ cm}}$$

$$A = \underbrace{2\pi r^2}_{\text{area of ends}} + \underbrace{2\pi r h}_{\text{lateral area}}$$

$$A = 2\pi r^2 + 2\cancel{\pi}\cancel{r} \cdot \frac{1000}{\cancel{\pi} r^{\cancel{2}}}$$

$$A = 2\pi r^2 + \frac{2000}{r}$$

$$A' = 4\pi r - \frac{2000}{r^2}$$

$$0 = 4\pi r - \frac{2000}{r^2}$$

$$\frac{2000}{r^2} = 4\pi r$$

$$2000 = 4\pi r^3$$

$$\frac{500}{\pi} = r^3$$

$$r = \sqrt[3]{\frac{500}{\pi}}$$

$$\boxed{r \approx 5.42\text{ cm}}$$

Notes:

If the function that you want to optimize has more than one variable, use substitution to rewrite the function.

If you are not sure that the extreme you've found is a maximum or a minimum, you have to check.

If the end points could be the maximum or minimum, you have to check.